TEAM FINANCE BROSITOS & BROSITAS · TUM QUANTUM HACKATHON 2026

Keep, review, or switch?

How a rule-keeping quantum circuit helps Germany's new pension fund decide, saver by saver



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Final pitch · 30 October 2026

THE CHALLENGE

Inflation is burning our savings

- Savings lose value every year
- Pensions alone will not be enough



THE DÖNER INDEX

€100 today buys about €76 of goods in 2035, and €48 in 2050

If prices rise 3% a year and savings earn 0%.



THE OPPORTUNITY

We saw an opportunity

- Germany is building a capital-funded pension
- Someone has to decide how to invest it



DISCLAIMER

We are students, not your financial advisers

This is a teaching model, not financial advice. If it were, our own portfolios wouldn't be this red. ⚠️

OUR CLIENT

The investment committee of Germany's new pension fund

2%
of gross wages go into a public fund

2028
the year it starts

0
guarantees: losses hit savers

Its independent committee decides how the money is invested. We help it with one decision.



THE CHALLENGE

Savers need growth, but cannot afford big losses

What savers want

- Growth that beats inflation
- Safety from big losses
- A sensible default if they never choose

Our questions

- Is tech worth the extra risk?
- Do losses stay within each saver's limit?
- Who protects people who never choose?

Takeaway Change a pension portfolio only when the evidence is strong.

Source: Alterssicherungskommission (June 2026)

THE SETUP

We decide one small slice: 3 of 6 sectors

90%

Core: a broad global index

We don't touch it.

10%

Satellite: 3 of 6 US sectors

Equal thirds of staples, manufacturing, energy, tech, health, utilities.

20

possible baskets

A laptop checks all of them: our exact answer key.

Takeaway A small, checkable decision. We claim no quantum advantage.

Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

THE SAVERS

Three savers, three different loss limits



LENA | Growth

“I can handle big swings.”

Loses more than 8% in at most 6 of 60 months



THOMAS | Balanced

“I never chose a profile.”

Loses more than 5% in at most 6 of 60 months



ANNA | Risk-sensitive

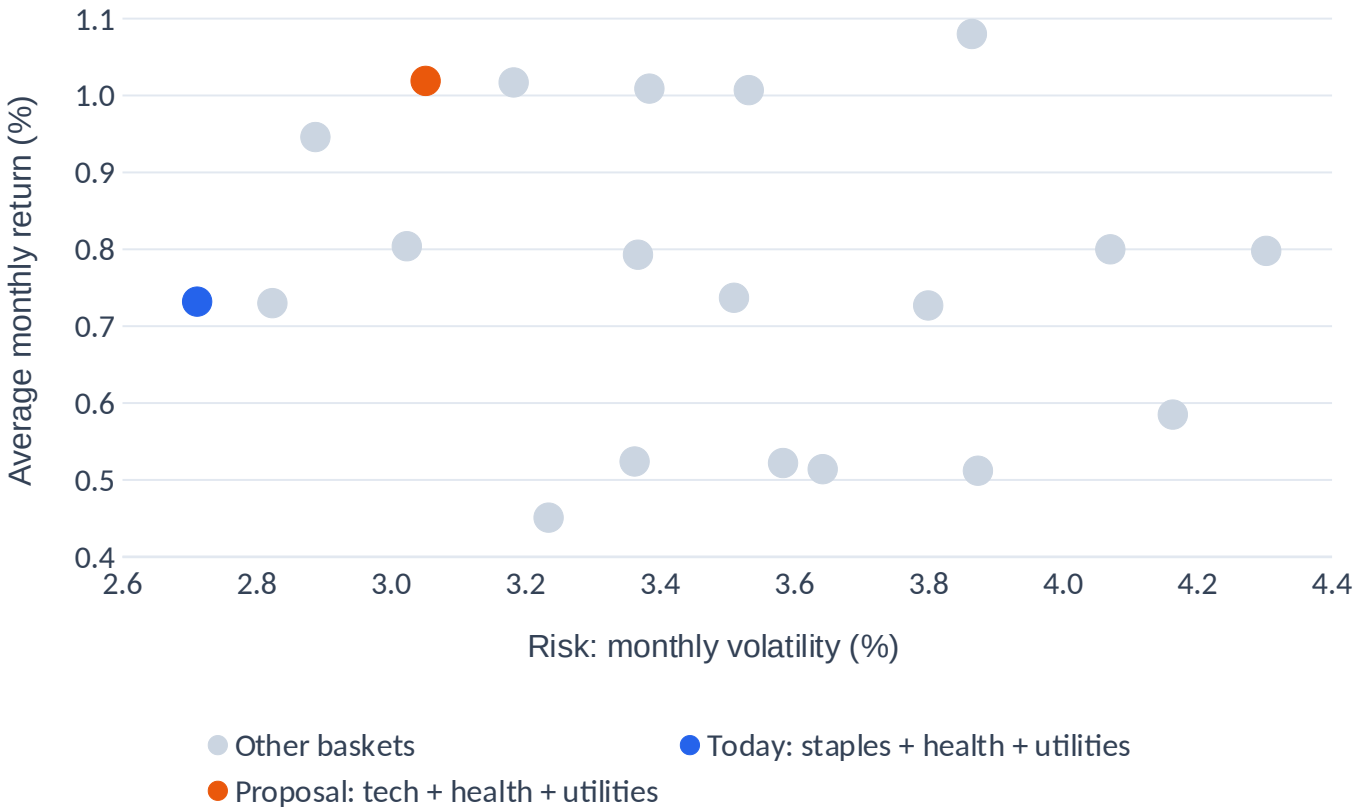
“I can't afford big losses.”

Loses more than 3% in at most 6 of 60 months

Takeaway Thomas never chose a profile, so he deserves extra care.

Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

Swap staples for tech: more return, but more risk



The proposal

- Return: 0.73% → 1.02% a month
- Risk: 2.71% → 3.05% a month
- Today's basket: lowest risk of all 20
- No return can buy a breach of a loss limit

Takeaway Is the extra return worth the extra risk? It depends on the saver.

Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

HOW WE DECIDE

Rules first, then the evidence decides

1

Check the loss limit

Baskets that break it are out. No return buys them back.

2

Score the rest

More return is good, more swings are bad, weighted per saver.

3

Switch only on strong evidence

Same winner in all 3 test settings and in at least 70% of 500 replays. Otherwise: review, and hold.

Takeaway The burden of proof is on the change.

Source: src/qhf/decision.py · 3 test settings: base case, tech earns 0.5 points less, sectors move more together

RESULTS

One proposal, three answers



LENA | Growth

SWITCH

Better in 89% of 500 replays

Wins every stress test



THOMAS | Balanced

REVIEW

Near-tie: flips under stress

Hold until the committee decides



ANNA | Risk-sensitive

KEEP

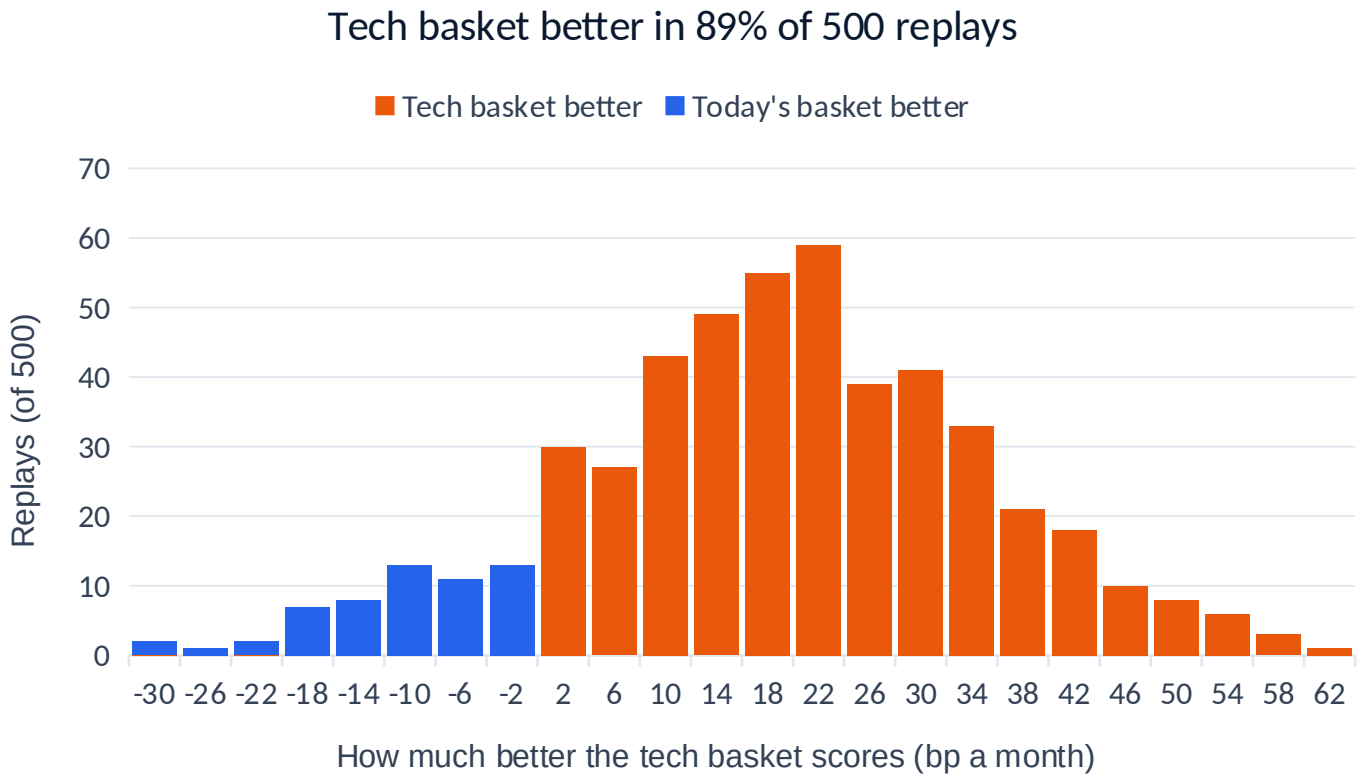
Lowest risk of all 20 baskets

Tech sits right on her loss limit

Takeaway “Review” is a valid answer, not a failure.

Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

Lena: switching to tech wins every test





Switch

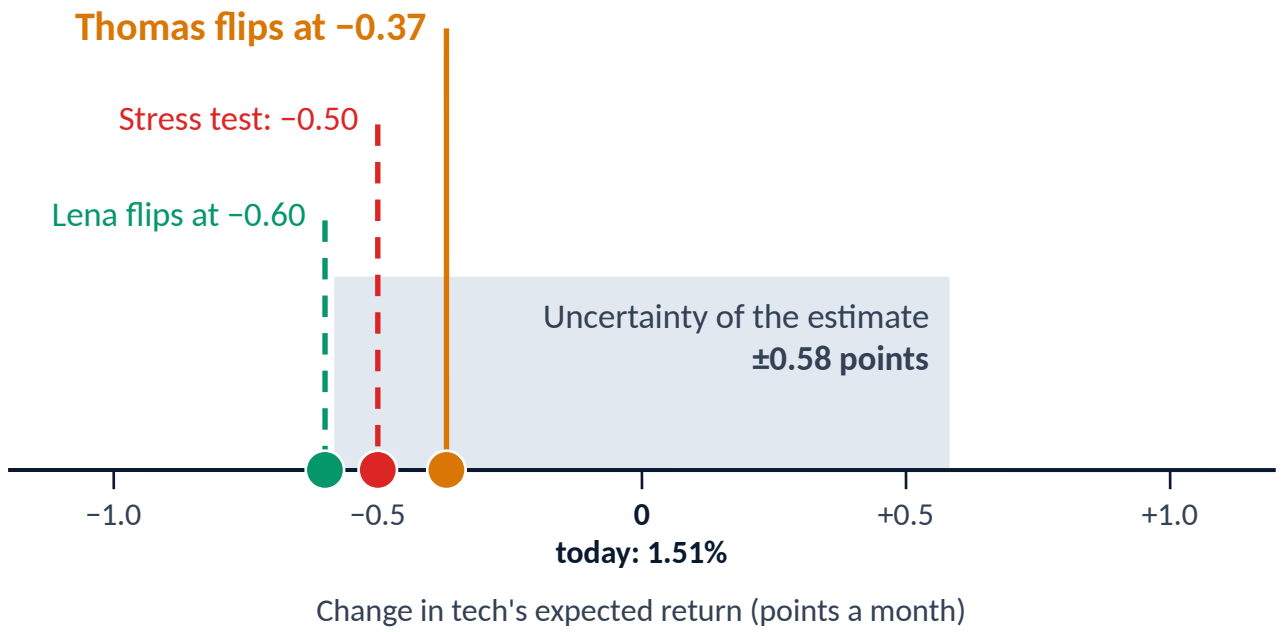
- +0.29 points return a month
- Better in 89% of 500 replays
- Wins every stress test
- One-off switching cost earned back within a month

Takeaway Strong evidence → switch.

Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

Thomas: the data cannot tell the two baskets apart

How much tech's monthly return may change before Thomas's answer flips





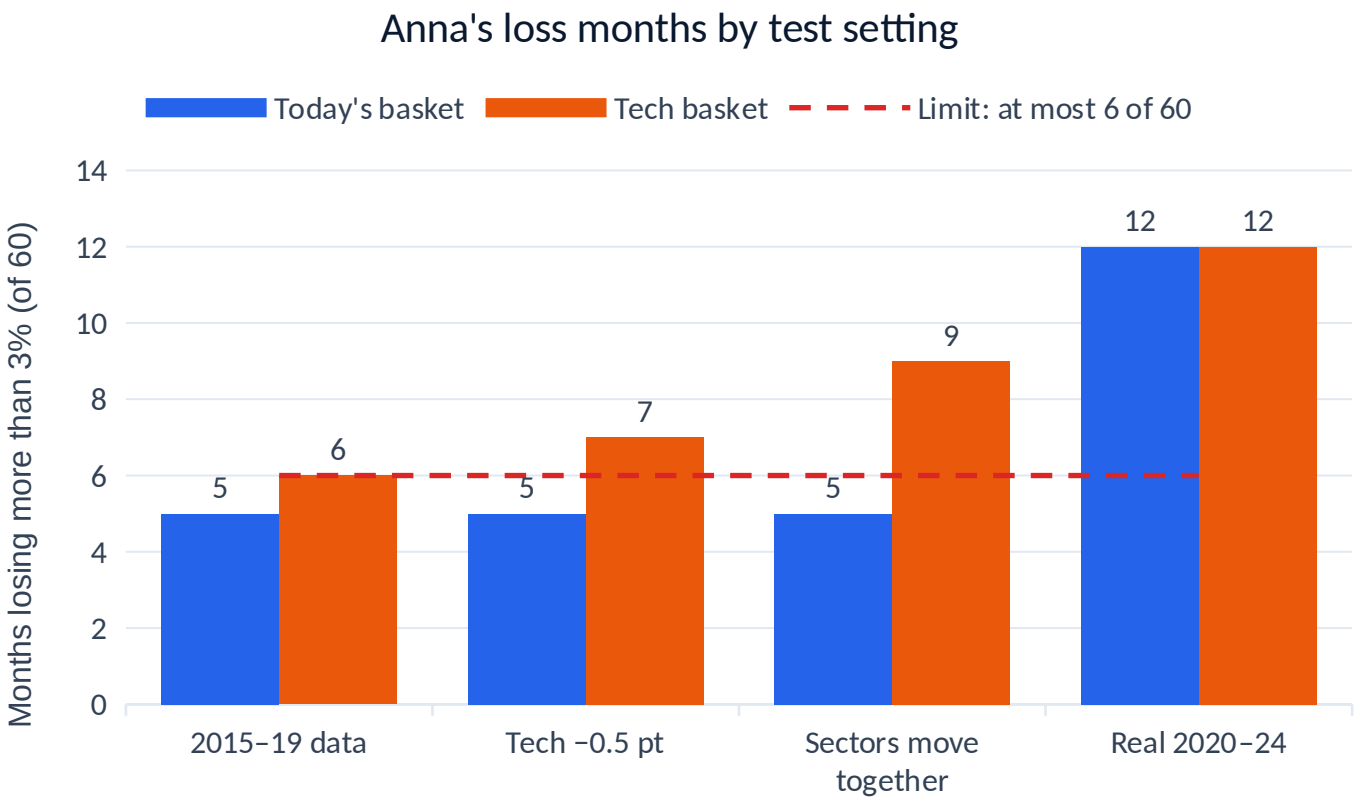
Review

- Lead: only 12 bp a month
- Flips if tech earns 0.37 less
- Estimate wobbles by ±0.58
- Stress test (−0.5) flips it

Takeaway Too close to call → hold, and let the committee decide.

Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

Anna: today's basket is the safer choice





Keep

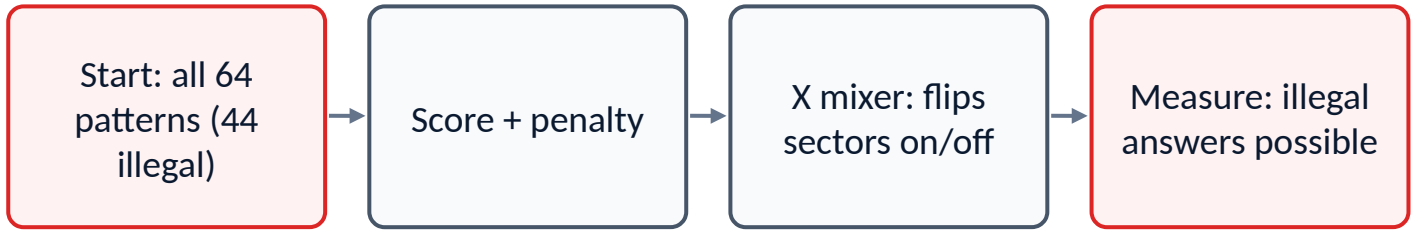
- Lowest risk of all 20 baskets
- Tech sits on her limit: 6 of 60
- Fails both stress tests: 7 and 9

Takeaway Safety first → keep.

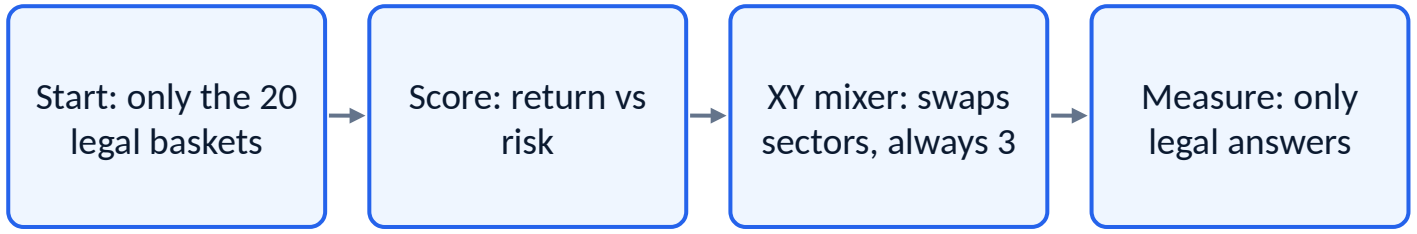
Source: our canonical run (results/canonical) · data: Kenneth French, 6 US industries, Jan 2015–Dec 2019

Our quantum circuit can only suggest legal baskets

Organisers' penalty circuit



Our rule-keeping circuit



Both repeat score and mix in 3 layers; a laptop tunes the dials.

Our idea

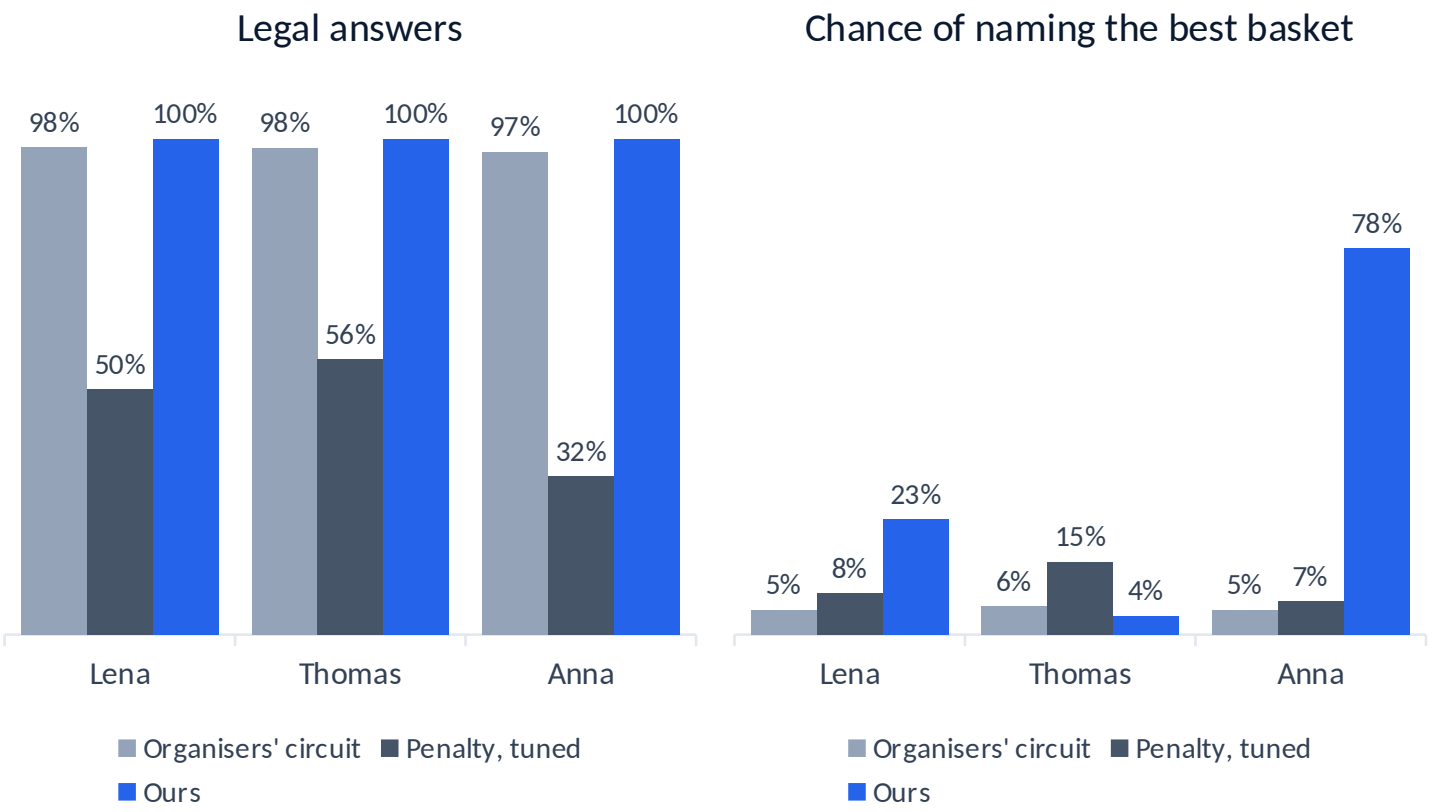
- Start with the 20 legal baskets only
- Mix by swapping sectors: always exactly 3
- Illegal baskets never appear

Takeaway The rule is built into the circuit, not punished afterwards.

Method: Hadfield et al. (2019), QAOA with Dicke start and XY mixer · code: src/qhf/circuits.py

QUANTUM RESULTS

Always legal, and far more accurate for Anna



100%
legal answers, every run

78%
Anna's best basket found (organisers': 5%)

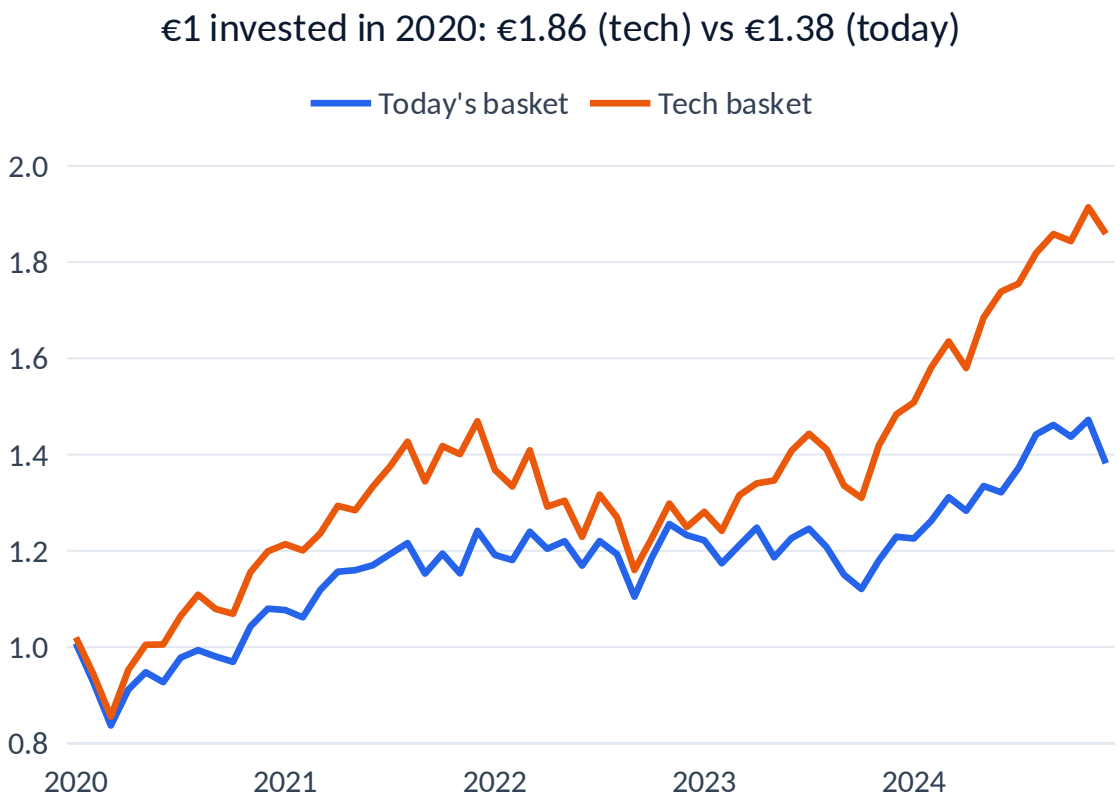
4%
for Thomas: no better than guessing

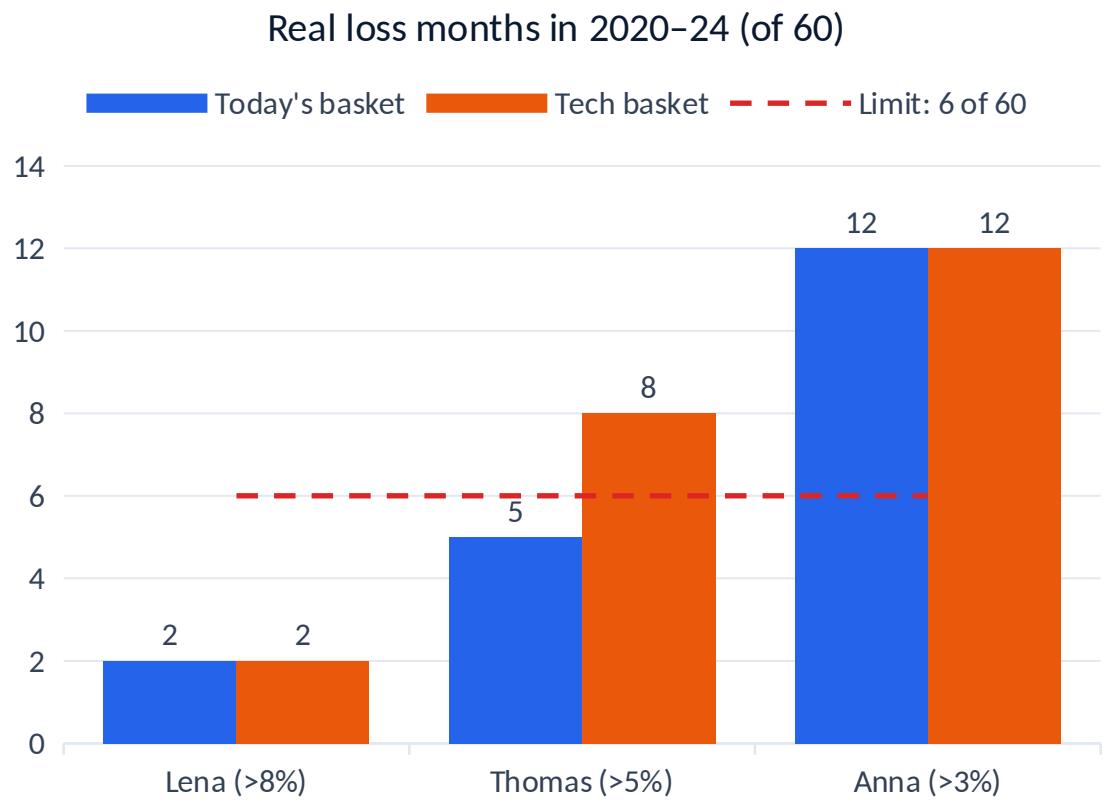
Takeaway A clear win where the answer is clear; no magic on a near-tie. No quantum advantage claimed.

Source: results/canonical/quantum_summary.csv · ideal simulation, 3 layers, median of 10 runs, 4,096 looks each

REALITY CHECK 2020–2024

Tech grew more, but losses came more often





Takeaway Hindsight only: Anna's limit broke for both baskets. Our loss model was too optimistic.

Source: results/canonical (holdout.csv, loss_limits.csv) · Jan 2020–Dec 2024, never used to decide

POLICY

Three rules for using this responsibly

1 REVISED

Protect the default

Savers choose a profile. Those who don't land in balanced and need protection from fragile bets.

2 ADOPTED

Law sets limits, committee sets the mix

Limits are checked exactly and never traded for return.

3 ADOPTED

Burden of proof

Switch only with the same winner in every test and at least 70% support.

Takeaway Policy sets the limits; people make the final call.

Sources: Alterssicherungskommission (June 2026); team policy brief

CONCLUSION

Clear answers, from a circuit that keeps the rules



Lena

SWITCH



Thomas

REVIEW



Anna

KEEP

No quantum advantage, and we say so. One command reproduces every number.

THANK YOU

Questions?

Code, decision log and every result: github.com/jonas221101/quantum-hackathon-finance

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